GATE

Computer Science & IT

Computer Networks

Application Layer

DPP: 1

Q1 Identify which name server(s) does not belongs to DNS hierarchy?
(A) Root Name Server
(B) Top Level Domain Name Server
(C) Real Name Server
(D) Authoritative Name Server

Q2 Consider the DNS resolver uses iterative query mechanism for resolution of the domain name "www.facebook.com". The number of DNS query-response pairs involved in completely resolving the domain name is _____ .

Q3 Which is/are valid HTTP request method?
(A) GET (B) PUT
(C) POST (D) HEAD

Q4 HTTP client send requests in non-persistent mode to HTTP server for a web page. The web page contains HTML file and one very small size image and one very small size audio file. The minimum number of TCP connections required to display the web page completely in your browser is _____ .

Q5 HTTP client send requests in persistent mode to HTTP server for a web page. The web page contains HTML file and three very small size images. The minimum number of TCP connections required to display the web page completely in your browser is _____ .

Q6 HTTP client send requests to HTTP server for a web page, identify the correct sequence in which the following protocols are used by the client host?
(A) HTTP, TCP, DNS (B) DNS, TCP, HTTP
(C) DNS, HTTP, TCP (D) TCP, DNS, HTTP

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Answer Key

Q1	(C)	Q4	3
Q2	3	Q5	1
Q3	(A, B, C, D)	Q6	(B)

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Hints & Solutions

Q1 Text Solution:

- Root Name Server **(Level-1)**
- Top Level Domain Name Server **(Level-2)**
- Real Name Server
- Authoritative Name Server **(Level-3)**

Q2 Text Solution:

Domain Name

$$\underbrace{\text{www.}}\quad \underbrace{\text{facebook}}_{\Uparrow\ \text{Authoritative name server}}\quad \underbrace{\text{com}}_{\Uparrow\ \text{com top level domain(TLD) name server}}\quad \Uparrow\ \text{Root Name Server}$$

$$\underbrace{\text{www.}}\quad \underbrace{\text{facebook}}_{\Uparrow\ \text{Authoritative name server}}\quad \underbrace{\text{com}}_{\Uparrow\ \text{com top level domain(TLD) name server}}\quad \Uparrow\ \text{Root Name Server}$$

Q3 Text Solution:

- GET
- PUT
- POST
- HEAD

Q4 Text Solution:

Non-Persistent HTTP

1. One TCP : HTTP Request for web page and base HTML file is transfered.

2. One TCP: HTTP Request for one object.

3. One TCP: HTTP Request for second object.

Q5 Text Solution:

Persistent HTTP:

1. One TCP: (connection established)

(a) HTTP request for web page, base HTML file is transferred

(b) HTTP request for object(s) on same TCP connection.

2. TCP connection close at end.

Q6 Text Solution:

DNS, TCP, HTTP

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